

URCHINS AND RELATIVES

Sea urchins, which belong to the class Echinoidea, gave their name to the whole phylum. The hard ossicles of an urchin are plates that interlock to form a shell called a test. The animal's spines help in locomotion and defence. The mouth generally faces downwards and the anus upwards – with rows of tube feet running between the two.

10–11 cm
4–4¼ in

INDO-PACIFIC SAND DOLLAR

Sculpsitechinus auritus
F: Astriclypeidae

This Indo-Pacific urchin of coastal sands is a typical member of a group of burrowing sea dollars, named for their flattened appearance.

4–5 cm
1½–2 in

BANDED SEA URCHIN

Echinothrix calamaris
F: Diadematae

The shorter spines of this Indo-Pacific reef-dwelling urchin are painfully venomous. Cardinalfish (Apogonidae) often hide among them for protection.

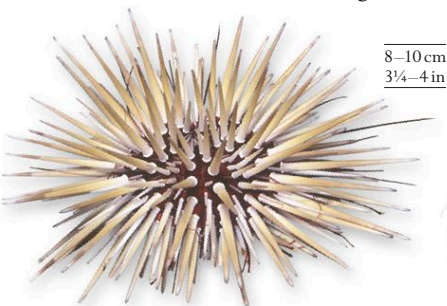
20 cm
8 in

RED URCHIN

Astropyga radiata
F: Diadematae

This Indo-Pacific lagoon species belongs to a family of tropical urchins with long hollow spines. It is often carried by a crab, *Dorippe frasnese*.

8–10 cm
3¼–4 in



MATHA'S URCHIN

Echinometra mathaei
F: Echinometridae

This Indo-Pacific species belongs to a group of urchins characterized by thick spines. It lives in crevices on coral reefs.

16–18 cm
6½–7 in



EDIBLE URCHIN

Echinus esculentus
F: Echinidae

This large globular urchin common on N.E. Atlantic coasts has a variable colour. It grazes on invertebrates and algae, especially kelp.

FIRE URCHIN

Asthenosoma varium
F: Echinothuriidae

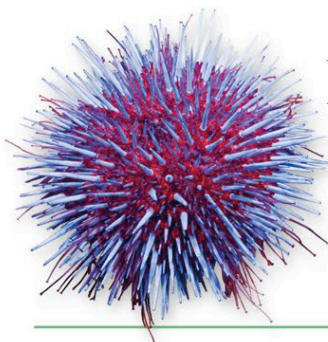
This somewhat flat Indo-Pacific species of sandy lagoons has a shell sufficiently flexible for the urchin to enter crevices. It can deliver a painful sting.

20–25 cm
8–10 in

flexible test



8–10 cm
3¼–4 in



PURPLE URCHIN

Strongylocentrotus purpuratus
F: Strongylocentrotidae

An inhabitant of the underwater kelp forests along the Pacific coast of N. America, this urchin is often used in biomedical research.

7–8 cm
2¾–3¼ in



SEA POTATO

Echinocardium cordatum
F: Loveniidae

Heart urchins, such as this globally widespread species, are detritus-feeding burrowers that lack the obvious radial symmetry of other urchins.

SEA CUCUMBERS

Sea cucumbers are soft tubular animals, with a mouth surrounded by a ring of food-collecting tentacles at one end, and an anus at the other. Some Holothuroidea shovel or burrow in sediments using their tentacles; others use rows of tube feet to crawl along the seabed.



YELLOW SEA CUCUMBER

Colochirus robustus
F: Cucumariidae

This echinoderm belongs to a family of thick-skinned sea cucumbers. Like this Indo-Pacific species, many are covered in knobby projections.

5–8 cm
2–3¼ in

15–18 cm
6–7 in

SEA APPLE CUCUMBER

Pseudocolochirus violaceus
F: Cucumariidae

Sea apples are brightly coloured, highly poisonous, reef-dwelling sea cucumbers. This species has yellow or orange tube feet, but variable body colour.



warning coloration

sticky mass ejected to repel predators



OCELLATED SEA CUCUMBER

Bohadschia argus
F: Holothuriidae

Distributed from the W. Indian Ocean around Madagascar to the South Pacific, this large species is variable in colour, but is always spotted.

38–60 cm
15–23½ in

25–30 cm
10–12 in



EDIBLE SEA CUCUMBER

Holothuria (Halodeima) edulis
F: Holothuriidae

Large sea cucumbers are especially abundant in tropical waters. This Indo-Pacific species is harvested and dried for human consumption.



60 cm
23½ in

VERMIFORM SEA CUCUMBER

Synapta maculata
F: Synaptidae

Typical of its family, this Indo-Pacific species is a soft-bodied, worm-like sea cucumber that burrows in soft sediment. It lacks tube feet.

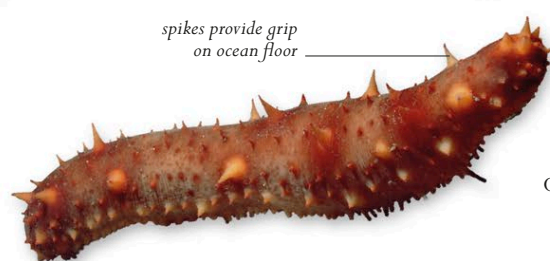
spikes provide grip on ocean floor

35–40 cm
14–16 in

GIANT CALIFORNIA SEA CUCUMBER

Apostichopus californicus
F: Stichopodidae

One of the largest sea cucumbers of N. America's Pacific coast, this fleshy spiked species is fished as a local delicacy.



60–75 cm
23½–30 in

PRICKLY REDFISH

Thelenota ananas
F: Stichopodidae

This echinoderm belongs to a family of sea cucumbers covered in fleshy spikes. It occurs on sandy seabeds around Indo-Pacific coral reefs.



30 cm
12 in

DEEP SEA CUCUMBER

Kolga hyalina
F: Elpidiidae

Virtually worldwide in distribution at depths of up to 1,500m (5,000ft), this is one of many little-known sea cucumbers found on ocean floors.